PlantSense

Evidence-backed plant reliability decisions

Manufacturing: Intelligent Operations & Industrial Efficiency | AI Builder Cup 2026

Prototype proposal and local validation draft. Cloud deployment and live Gemini validation remain pending authentication.

The problem

Plants can detect an abnormal signal without quickly knowing its cause, downstream consequences, or whether maintenance should interrupt production. Technical knowledge, production commitments and repair history are usually reviewed separately.

The solution

PlantSense links a spatial plant view to one complete reliability workflow: persistent detection, evidence gathering, uncertain diagnosis, deterministic intervention economics, human approval, repair verification and reusable confirmed knowledge.

PlantSense / OPERATIONS

SIMULATED DEMO PLANT · API connected · NP

WORKSPACE > NARMADA PRECISION WORKS

Plant overview

REPLAY

A clear view of your plant. An informed next move.

Shift A · 0 simulated min

ASSET HEALTH	PLANT OUTPUT	EXPECTED OPTION COST	VERIFIED REPAIRS
12 / 12 healthy	200 parts / h	—	0 confirmed cases
All systems within normal state	Nominal capacity 200 parts / h	Configured shift economics	Evidence retained for the next incident

Plant assets

12

LINE A

- P-107 Hydraulic pump A
- M-107 Drive motor A
- PR-101 Hydraulic press A
- CV-101 Conveyor A

LINE B

- P-207 Hydraulic pump B
- M-207 Drive motor B
- PR-201 Hydraulic press B
- CV-201 Conveyor B

UTILITIES

- CL-001 Cooling unit
- AC-001 Air compressor
- EL-001 Electrical supply

DISPATCH

- FG-001 Finished goods

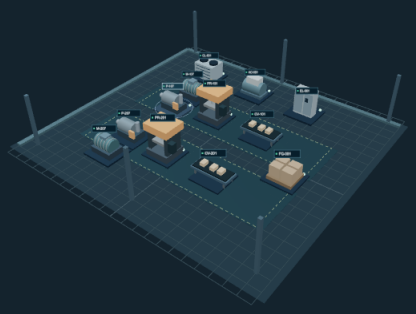
Healthy · Attention

Digital plant / 3D view

2D plan

NARMADA PRECISION WORKS

Hydraulic forming · 2 production lines



Healthy · Degrading · Downstream impact

Telemetry replay

UCI measured replay

12×

Explore a scenario: Pump degradation

Start scenario

P-107 / Hydraulic pump A

Healthy

Supplies pressure and flow to Press A

PRESSURE	FLOW	TEMPERATURE
161.01 bar	6.7 L/min	36.26 °C
Awaiting replay samples	Awaiting replay samples	Awaiting replay samples
UCI cycle #1788	UCI cycle #1788	UCI cycle #1788

POWER	VIBRATION
2,539.07 W	0.55 mm/s
Awaiting replay samples	Awaiting replay samples
UCI cycle #1788	UCI cycle #1788

Activity log

SESSION AUDIT

Start a scenario to observe the reliability workflow.

Plant intelligence

EVIDENCE → DECISION → OUTCOME

Investigation · Decision · Work order · Memory

Your plant is in view.

Start a scenario. PlantSense will detect a persistent change and investigate it automatically.

Detect → Investigate → Decide

Whole-plant state and dependency effects are simulated. Measured pump replay is attributed separately.

Ask the plant

5 left

What is affecting Line A?

System online · isolated demo session

PlantSense / Evidence-backed reliability

What the working prototype demonstrates

• Twelve selectable assets, two production lines, shared utilities and downstream impact paths.

• Public hydraulic measurements replayed through a fitted normal baseline and Isolation Forest.

• Inspectable telemetry and cited demo procedures; conflicting-sensor uncertainty is explicit.

• Four actions compared over the same eight-hour horizon using integer-paise costs and labelled assumptions.

• Versioned human approval, technician correction and three-window recovery verification.

• A later incident retrieves the verified repair while retaining the original diagnosis.

Google AI and Cloud implementation

The cloud integration uses ADK/Gemini for tool-based investigation; Cloud Run for web/API/private worker; Cloud Tasks for authenticated asynchronous execution; Firestore for transactional workflow state and vector retrieval; Firebase anonymous identity; and Cloud Storage for source artifacts. Deployment scripts and locked container builds are supplied. The local investigator is explicitly scripted and does not substitute for live-model validation.

Trust and differentiation

The differentiator is the coherent decision loop: shared operational evidence, competing maintenance and production objectives, transparent arithmetic, human control, and proof of repair. The scene communicates where consequences occur. No automatic real-plant control, certified safety behaviour, online retraining or zero-configuration factory integration is claimed.

Evidence and limits

The public dataset has 2205 cycles and only 10 fully healthy stable cycles. The detector fits 6 healthy cycles. The held-out subset has 38 cycles, including only 1 healthy cycle. Reported precision/recall on that subset are 1.00/1.00; this small, constrained evaluation does not establish industrial generalization or robust false-alarm performance.

Software verification covers ownership, stale approvals, idempotency, deterministic costs, failed recovery and memory. Browser tests exercise five complete workflows. Actual live-model diagnosis, grounding, abstention and tool performance still require evaluation on the deployed project.

Business value

PlantSense makes the cost and operational consequence of intervention alternatives inspectable. Configured contribution margin, downtime, penalties and failure probabilities demonstrate decision mechanics; they are not customer savings or calibrated risk estimates. An industrial pilot would measure decision time, confirmed diagnostic accuracy, verified repair rate, avoided output loss and operator acceptance.

Delivery and expansion

The submission focuses on the full incident-to-resolution loop. Future work adds qualified site-specific manuals, more healthy operating regimes, actual plant adapters, broader asset models, and CMMS integration. The canonical model separates plant configuration from reusable reasoning and workflow logic.

Source attribution

Helwig, Pignanelli and Schuetze (2015), Condition monitoring of hydraulic systems, UCI Machine Learning Repository, DOI 10.24432/C5CW21, CC BY 4.0. Cycle means, filtered/split evaluation and reordered replay are derived transformations. Factory geometry and demo SOPs are original synthetic material.